

Jerry Liu

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EDUCATION

University of Waterloo

Waterloo, Ontario

BASc, Mechatronics Engineering

2025 – 2030

- **Awards:** Arthur F. Church Entrance Scholarship (\$10,000) – Awarded to only 1 student in the Mechatronics cohort for outstanding academic and extracurricular achievement.

SKILLS

Languages: TypeScript · Python · JavaScript · SQL · C++

Frameworks: React · Next.js · NestJS · Node.js · Prisma · FastAPI · Mapbox GL JS

Databases: PostgreSQL · PostGIS · Redis/BullMQ · Supabase

AI: PyTorch · Transformer Models · Embeddings · LLM Vision Extraction · Gemini API

Infrastructure: Docker · AWS S3 · GitHub Actions · Git · Playwright

RELEVANT EXPERIENCE

Full-Stack Software Engineering Intern (member of the founding team)

Remote

Parcyl | [Link](#) | Site intelligence platform for real estate diligence; 2-person engineering team

May 2026 – Aug 2026

- Built a LangGraph multi-agent loop that drives the product as a synthetic user, finds defects, and ships verified fixes as draft PRs
- Designed and built an agentic report engine: 11 ReAct agents generating research reports for any property on demand, each personalized to the user's portfolio, uploaded documents and prior deals
- Built an async document pipeline on S3 and BullMQ that turns PDFs, spreadsheets and scans into grounded context for the chat and report agents, cited by quote and page
- Built a feedback-triage agent that deduplicates user feedback in three stages and posts a batched digest carrying the underlying evidence directly to Slack

Undergraduate Research Assistant

Waterloo, Ontario

University of Waterloo

Feb 2026 – June 2026

- Designed drone-based remote sensing platform for greenhouse gas detection using mid-infrared spectroscopy
- Developed multi-sensor fusion pipeline integrating LiDAR, GPS/IMU, and anemometer data for source localization
- Conducted literature review and contributed to component selection under Prof. Shunde Yin

Research Assistant

Waterloo, Ontario

University of Waterloo

August 2024 – June 2025

- Researched and selected multi-modal sensors for time-series degradation analysis on CNC milling machines
- Developed Python data processing pipelines for data trials, converting raw sensor data to usable PyTorch files
- Collected and processed high-frequency time-series data for remaining-useful-life prediction research using transformer models

Co-Founder, Logistics Lead

Waterloo, Ontario

NeoDev League | [Link](#)

May 2024 – June 2025

- Co-founded a non-profit organizing large-scale engineering competitions for high school students
- Led logistics for a 10-hour hackathon-style event, managing scheduling, facilities, and technical resources
- Secured \$10,000+ in sponsorships and backed by convictional(yc w19); reached 150 students across 10 schools

PROJECTS

Canopi: Rental Matchmaker | [Link](#)

March 2026

- Developed full-stack rental platform with Next.js, TypeScript, and Supabase, integrating a custom scraper to aggregate real-time housing data across Canada
- Implemented a custom compatibility algorithm to calculate user-to-property match scores and integrated the Gemini API to build an AI chat interface for inferring user preferences through natural language.
- Posted project on LinkedIn, where it drew inbound interest from a founder and led directly to the Parcyl internship

Neural Chess Engine | [Link](#)

Nov 2025

- Built and trained a 6-layer PyTorch neural network using a custom 12-channel (12×8×8) board encoding on millions of high-ELO games
- Implemented a minimax engine with alpha-beta pruning and deployed the model to Lichess via the lichess-bot API for real-time play

Firefighter Robot | [Video](#)

Feb 2025 – June 2025

- Programmed autonomous navigation and fire detection in GCBASIC, integrating wall sensors, phototransistors, and IR flame sensors for real-time control
- Designed and manufactured a custom PCB and 3D-printed chassis; assembled, soldered, and debugged the full system to validate end-to-end performance